Unit 1: Exercises

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01 • Read conditions and nonlinearity

A device remains in the same state during two same-polarity readings: 2 µA at 0.1 V and 6 µA at 0.2 V. Calculate apparent conductances and test a linear, origin-passing resistor model. What conditions should accompany a resistance measurement?

E02 • Operations and polarity

Define SET, RESET, forming and read. In a specified bipolar device, a positive top-electrode voltage relative to the bottom electrode performs SET. Which polarity is used for RESET under this operating convention? Can it be generalized to other electrode stacks? Why limit programming current?

E03 • Evidence and mechanisms

A metal/oxide/metal device shows reproducible bipolar hysteresis. Does this alone establish ECM or VCM? Identify possible mobile species and two types of distinguishing evidence, with limitations.

E04 • Ideal crossbar

Let row voltages be $V = (0.2, 0.1)$ V and conductances $G_{11} = 2$, $G_{12} = 1$, $G_{21} = 1$, $G_{22} = 3$ µS. Assume linear devices, zero wire resistance and virtual-ground columns. Find both column currents and every branch current. If wire drop instead leaves 0.08 V at every crosspoint in row 2, with other conditions unchanged, recalculate and explain whether the original ideal formula still applies.

E05 • Read margin

At a specified operating point, HRS currents range from 1 to 3 µA and LRS currents from 5 to 9 µA. Define margin as minimum LRS current minus maximum HRS current. Calculate it and give thresholds that distinguish the closed intervals using the rule “strictly greater than threshold means LRS.” If newly observed distributions overlap, is an error rate determined by this information alone?

Unit 1: Exercises (continued)

E06 • Reliability claims

Two tests report 10^6 cycles and ten years of retention. List at least four conditions required for comparison, distinguish cycle-to-cycle and device-to-device variability, and explain why one hour at 85 °C alone does not establish ten years at room temperature.

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https://github.com/xinhi6716/resistive-memory-self-study